

EDUCATION

University of California, Berkeley Philosophical Doctorate, Organic Chemistry Advisor: Professor Richmond Sarpong	August 2020 – May 2025 (Anticipated) GPA: 3.95/4.00
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George Washington University Bachelor of Science in Chemistry, ACS Certified Biochemistry concentration, Biology minor Advisor: Professor Cynthia Dowd	August 2016 – May 2020 GPA: 3.95/4.00 Major GPA: 3.98/4.00
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EXPERIENCE

University of California–Berkeley Department of Chemistry <i>PhD Candidate, Dr. Richmond Sarpong</i>	Berkeley, CA October 2020 – Present
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- Developed a method for pyrimidine-to-pyrazole ring contraction by formal carbon deletion
- Developed a method for ¹⁵N core-labeling of nitrogen heteroaromatics
- Developed four generations of new organic halogenation reagents
- Developed a method for the transaminative synthesis of *N*-malonyl azinium ylides
- Discovered a Lewis-acid catalyzed, inverse electron demand 1,3-dipolar cycloaddition reaction of *N*-malonyl azinium ylides
- Planned and worked toward the enantioselective total synthesis of a tetracyclic sesquiterpenoid natural product
- Investigated a method for visible-light mediated N-atom transposition of pyrazoles and indazoles
- Prepared an Organic Syntheses protocol for the ring contraction of pyrimidines to pyrazoles
- Built and maintained HPLC and HeptaOchem photobox systems; maintained GCMS for routine analysis

Collaborative Work

- **With the Sigman Group, U. Utah:** performed MLR and classification analysis for predictive model development
- **With the Coley Group, MIT:** consulted for retrosynthetic prediction software for natural products synthesis
- **With the Paton, Sigman Groups; CSU, U. Utah:** developed a cheminformatic platform the analysis of skeletal edits
- **With the Grzybowski Group, UNIST-gil:** evaluated a programmatic platform for multidirectional reaction prediction

Teaching Experience

- Graduate Student Instructor for **CHEM 12A** and Lab (Organic Chemistry for Chemistry Majors, Prof. Alanna Schepartz)
- Graduate Student Instructor, Lecturer, and Course Design for **CHEM 115** (Advanced Organic Chemistry, Prof. Dean Toste)
- Mentored one junior graduate student, two visiting students, and one undergraduate

George Washington University Department of Chemistry <i>Undergraduate Researcher, Dr. Cynthia Dowd</i>	Washington, DC December 2017 – May 2020
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- Synthesized a large library of fosmidomycin derivatives as antimalarial and antitubercular agents
- Investigated the bioactivity of the derivatives (and prodrugs thereof) as bisubstrate inhibitors of *Dxr*
- Developed a short, modular synthesis for the derivatives

Vertex Pharmaceuticals, Inc. <i>Process and Cheminformatics Intern, Dr. Stefanie Roeper and Dr. Rebecca Swett</i>	Boston, MA May 2020 – August 2020
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- Computationally evaluated the Buchwald-Hartwig amination in the synthesis of a stage 2 API
- Parameterized ligands for multiple and non-linear regression analysis
- Built Python and Excel packages for data analysis distributed interdepartmentally

<i>Process Chemistry Intern, Dr. Stefanie Roeper</i>	June 2019 – August 2019
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- Developed, executed, and optimized the synthesis of a polysubstituted pyrroloindazole Stage 2 API
- Developed chromatography-free protocols key intermediate isolations
- Worked closely with chemical engineering interns to complete joint projects

G. LOGAN BARTHOLOMEW

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PUBLICATIONS

- **Bartholomew, G. Logan**, Kraus, Samantha L.; Karas, Lucas J.; Carpaneto, Filippo; Bennett, Raffael; Sigman, Matthew S.; Yeung, Charles S.; Sarpong, Richmond. "[¹⁴N to ¹⁵N Isotopic Exchange of Nitrogen Heteroaromatics through Skeletal Editing](#)" *J. Am. Chem. Soc.* **2024**, *146*, 2950–2958.
- **Bartholomew, G. Logan**, Carpaneto, Filippo; Sarpong, Richmond. "[Skeletal Editing of Pyrimidines to Pyrazoles by Formal Carbon Deletion](#)" *J. Am. Chem. Soc.* **2022**, *144*, 22309–22315.

PRESENTATIONS

- **Poster Presentation**, “New Methods for Core Atom Transformations of Nitrogen Heteroaromatics”, Gordon Research Conference – Heterocycles, Salve Regina University, Newport, Rhode Island, 2024.
- **Oral Presentation** with Jihye Roh (Coley lab, MIT), “Higher-Level Computer-Aided Retrosynthesis for Complex Molecules”, NSF Center for Computer Assisted Synthesis Annual Meeting, UCLA, Los Angeles, California, 2024.
- **Oral Presentation**, “New Methods for Core Atom Transformations of Nitrogen Heteroaromatics”; **(a)** College of Chemistry Symposium Day, University of California–Berkeley, 2024; **(b)** Corteva Agriscience Chemistry Scholars Symposium, Indianapolis, Indiana, 2024; **(c)** Gordon Research Seminar – Heterocycles, Salve Regina University, Newport, Rhode Island, 2024.
- **Poster Presentation**, “¹⁴N to ¹⁵N Isotopic Exchange of Nitrogen Heteroaromatics through Skeletal Editing”, **(a)** Merck Pharmaceuticals & Gilead Sciences Days, University of California–Berkeley, 2024 **(b)** Bay Area Chemistry Symposium, University of San Francisco, 2023. **Awarded Best Poster.**
- **Poster Presentation**, “Formal Single Atom Manipulations for Pyrimidine Skeletal Editing”, **(a)** Gilead Sciences Day, University of California–Berkeley, 2023; **(b)** Merck Pharmaceuticals Symposium, University of California–Berkeley, 2022; **(c)** National Organic Symposium, San Diego, California, 2022. **Awarded Best Poster.**
- **Virtual Presentation**, "MaxBridge: A Tutorial for Network Guided Analysis of Complex Polycyclic Natural Products", Center for Computer Assisted Synthesis, 2022.
- **Oral Presentation**, “New Reagents for Asymmetric Catalytic Halogenation”, Graduate Research Seminar, University of California–Berkeley, 2021.
- **Virtual Presentation**, "A Computational Investigation of the Buchwald-Hartwig Amination in the synthesis of [a Stage 2 API]", Department of Chemical Development, Vertex Pharmaceuticals, 2020
- **Undergraduate Thesis Defense**, “Design and Synthesis of Fosmidomycin Analogues as Bisubstrate Dxr Inhibitors to Combat Malaria and Tuberculosis”, Department of Chemistry, George Washington University, 2020.
- **Virtual Presentation**, "MEPicides; Natural Product Analogues as Potent Malaria Combatants", SciMeetings: Virtual Symposium, American Chemical Society Spring National Meeting and Expo, 2020
- **Oral Presentation**, “Novel Small Molecule Design and Synthesis to Selectively Inhibit Dxr and Combat Malaria and Tuberculosis”, Gulf Coast Undergraduate Research Symposium, Rice University, 2019. **Awarded Outstanding Presentation in Biomedical Chemistry.**
- **Poster Presentation**, "Developing a Robust Synthetic Strategy to Prepare Poly-Substituted Indole Derivatives", Department of Process Chemistry, Vertex Pharmaceuticals, 2019.
- **Poster Presentation**, “Novel Small Molecule Design and Synthesis to Selectively Inhibit Dxr and Combat Malaria and Tuberculosis”, Research Days, George Washington University, 2019; **(b)** Undergraduate Research Symposium, George Washington University 2019; **(c)** Frontiers in Chemistry and Biology Interface Symposium, National Institutes of Health, 2019.

RECOGNITIONS

- **NSF Graduate Research Fellowship**, National, 2021
- **ACS Division of Organic Chemistry Award**, University, 2020
- **Alpha Chi Sigma Prize; Highest Academic Record in Chemistry Courses**, University, 2020
- **American Chemical Society Prize; Division of Analytical Chemistry**, National, 2020
- **American Institute of Chemists Prize; Graduating Undergraduate Student**, National, 2020
- **Biomedical Chemistry: First Prize**, Gulf Coast Undergraduate Research Symposium, 2019
- **Enosinian Scholarship**, Honors Thesis Fellowship, GWU, 2019
- **Goldwater Scholarship**, Senator Barry Goldwater Foundation, National, 2019
- **A.D. Britt Scholarship**, Research Fellowship, GWU, Awarded both 2018 and 2019
- **Analytical Chemistry Award**, National Academic Recognition, American Chemical Society, 2019
- **Presidential Scholarship**, 4 Year Academic Merit Scholarship, GWU, 2016

ACTIVITIES

Merck Disruptive Chemistry Program <i>Academic/Industrial Collaboration</i>	Berkeley, CA January 2023 – Present
Center for Computer Assisted Synthesis (CCAS) <i>Nationwide Symposium of Interdisciplinary Scientists</i>	Berkeley, CA October 2021 – Present
Bay Area Scientists in Schools (BASIS) <i>Outreach Program for Scientific Education</i>	Berkeley, CA January 2021 – Present
George Washington Undergraduate Review <i>Annual Undergraduate Research Publication</i> Founder, Managing Editor	Washington, DC January 2018 – Present

SKILLSET

Chemistry

- Organic reaction setups, Schlenk and glovebox techniques; manual and automated flash column chromatography, crystallization, recrystallization, manual chiral separation, and preparative HPLC setup and analysis
- HPLC setup, analysis, and chiral separations, LCMS setup and analysis, GCMS analysis, HRMS
- NMR (^1H , ^{13}C , ^{15}N , ^{19}F , ^{31}P , HSQC, HMBC, COSY, NOESY, DEPT, variable temperature, real-time NMR kinetics)
- Large scale reactions, semi-automated synthesis process development, high-throughput experimentation

Computational Chemistry and Data Science

- iQMol, QChem, Gaussian 16, Schrödinger Maestro (MacroModel, Jaguar, Glide), PyMol, and Mercury
- Standard DFT optimizations, excited state calculations, and transition state identifications
- Python proficiency (sklearn, RDKit, SciPy, datamol, numpy, and pandas modules)
- Git and GitHub